



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : **SECD2613 System Analysis and Design**
Task : **Assignment 4**
Title : **Group Assignment 4 (DFD 5%)**
Due : **27/4/2024**
Submission : **Submit softcopy (in .pdf format) via e-learning**

INSTRUCTIONS:

1. 3 ~ 4 students per group
2. All answers should be printed on and uploaded via link specified in e-learning.
3. Any plagiarism – copying directly from other groups will result in both groups will penalized by getting an 'F' grade.

Question 1:

Fruity Fresh Juice is a family food business which cooks and sells a wide variety of dishes. Fruity Fresh Juice wants to develop a system to facilitate the process of ordering their meals. A customer can order from Fruity Fresh Juice by calling them on the phone. Then Fruity Fresh Juice will ask the customer to give their information (phone number, the name, address), and order details. Once the order is accepted, the total, including tax and delivery, will be calculated. Then a receipt will be printed and sent to the customer and the order will be sent to the kitchen. After the meal is ready, it will be sent to the driver together with customer information. Then the driver will deliver the meal to the customer. To monitor the performance of the Fruity Fresh Juice, every customer order receipt will be sent to the management to prepare and print periodic performance reports by referring to previous order data.

- a. Draw a context-level data flow diagram for Fruity Fresh Juice. **(5 marks)**
- b. Draw the Level 0 Diagram for the Context Diagram developed in (a). **(10 marks)**
- c. Construct a child diagram for the **“Prepare and print Periodic performance Report”** process. **(5 marks)**

Question 2:

PrintIt Quick is a local printing service that specializes in quick and efficient printing for customers who need everything from business cards to flyers to banners. They wish to establish a basic but efficient system to handle orders and customer service. Customers visit PrintIt Quick to request printing services. They provide their printing requirements, such as document type, paper quality, and quantity, directly to the service assistant at the counter. The assistant notes down the customer's details, including their name and contact number, and then provides a cost estimate based on the printing options chosen. If the customer agrees to the price, they pay upfront, and the assistant issues a receipt with an order number. The order details are then passed to the printing team, who prepare and print the materials according to the customer's specifications. When the print job is completed, the customer is contacted via phone call or text message, using the contact number provided, to collect their order. To monitor the business's efficiency, PrintIt Quick maintains a logbook that records all transactions, including customer details, order specifics, and payment received. At the end of each week, the business owner reviews the logbook to assess how many orders were completed and to identify any trends in customer requests. This simple system allows PrintIt Quick to keep a personal touch with their customers while ensuring that all orders are tracked and fulfilled promptly.

- a. Draw a context-level data flow diagram for Fruity Fresh Juice. **(5 marks)**
- b. Draw the Level 0 Diagram for the Context Diagram developed in (a). **(10 marks)**

- c. Construct a child diagram for the “**Order Completion and Notification**” process.
(5 marks)